# Digital Existence as Ontological Intervention:

## Backstage Networks in Postwar Taiwanese Popular Music

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## 0 THE CLAIM

### THE PROBLEM

The history of popular music is written around front-stage stars, and MIR has inherited that bias. The arrangers, session musicians, and engineers who built the sound are absent from our datasets. **For them, absence from metadata means erasure from history.**

### THE CLAIM

Encoding these workers into a database is an **ontological intervention**. It gives them a digital existence they never had. Two independent methods, a collaboration network and MERT audio embeddings, then point to the same year, **1978**.

## 1 WHAT WE BUILT: PASMA

The Private Archive of Scanned Music Artifacts, reconstructed from vinyl liner notes by an LLM pipeline with an expert reviewing every stage.

**1,122** vinyl liner notes

**1957–1991** years covered

**3,688** people and institutions

**15 + 5** role categories (persons + institutions)

### THE PIPELINE, 5 STAGES

1 OCR correction
↓
2 Credit extraction
↓
3 Year cross-check
↓
4 Entity disambiguation
↓
5 Role classification

Every stage is checked and corrected by a domain expert.

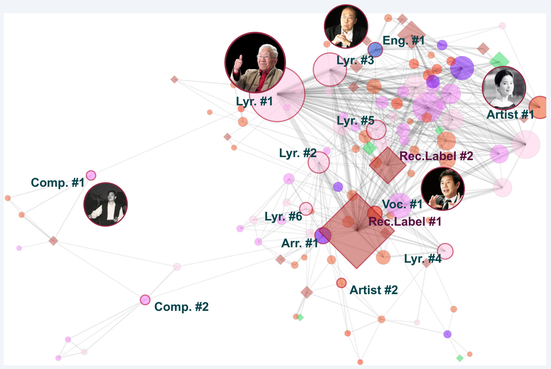
### DATA & DOCUMENTATION

**bkwmt.github.io/PASMA**

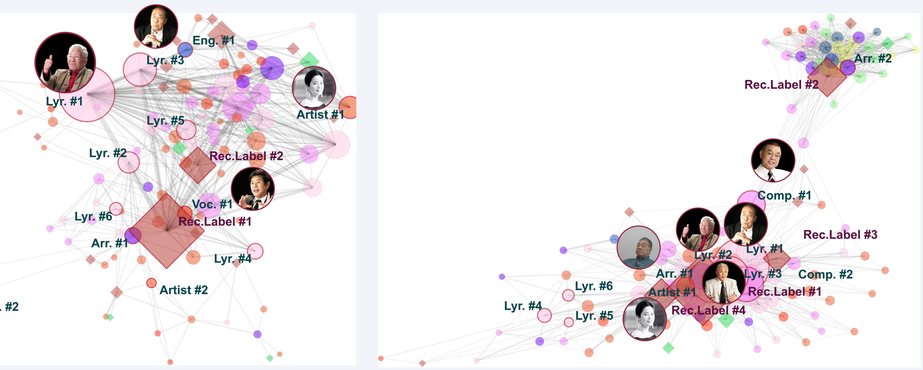
## 2 THE NETWORK TURNED AROUND 1978

Credits projected into a collaboration network. Circles are persons, diamonds institutions; colour is role. Each panel is cumulative.

Portraits mark people who also appear in our oral-history archive.




1957–1976 · Fragmented. Record labels at the centre; lyricists are the bridges; the Taiwanese-language community stands apart.




1957–1978 · The threshold. A cohort of musicians from the American military clubs appears, not yet absorbed.

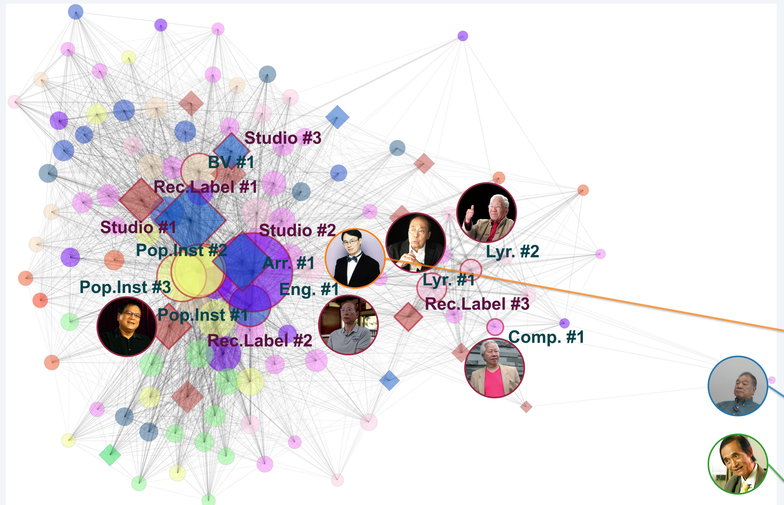
### THREE METRICS TURN TOGETHER

Clustering coefficient falls from **0.75** (1960) to below **0.05** after 1978.

Largest component crosses **0.95** of all actors around 1976 and stays there.

Louvain modularity drops from **0.69** (early 1970s) to **0.51** (1980).

**1976–1978 is not a chosen cut-off. It is a structural threshold that coincides with the adoption of multitrack recording in Taiwan.**




1957–1991 · One giant component. Recording studios sit at the centre; labels and lyricists are pushed to the edge.

Coloured lines anchor the four arrangement traditions of panel 4 to their arrangers in this network.

Performing Artist · Vocalist · Popular Instrument · Classical/Trad. Instrument
Backing Vocal · Arranger · Composer · Lyricist
Recording Engineer · Producer · Recording Studio · Record Label
Ensemble · Top-N (Person) · Top-N (Institution)

## 3 WHO BRIDGES THE NETWORK CHANGED

Role enrichment among the top 30 bridging actors (lowest Burt constraint). Above 1 means overrepresented; "—" means absent from the top 30.

| Role | '57–'76 | '57–'78 | '57–'91 |
|---|---|---|---|
| Lyricist | 1.63 | 0.69 | 0.53 |
| Vocalist | 0.32 | 0.31 | — |
| Arranger | 0.83 | 2.78 | 3.06 |
| Pop instrumentalist | — | 3.16 | 1.89 |
| Backing vocal | — | — | 4.01 |
| Recording engineer | 5.83 | 5.00 | 1.57 |

Lyricists and singers fall below parity; arrangers, players, backing vocalists, and engineers take the bridges.

**Singing stars were never connected this industry.**

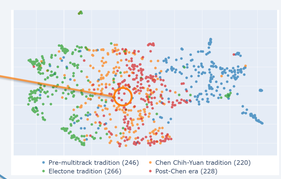


### THE STRONGEST BRIDGE

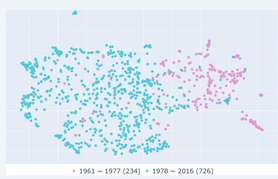
Chen Chih-Yuan ("陳志遠", 1941–2011), arranger. Credited on **178** albums, connected to **338** collaborators; the lowest constraint in the whole 1957–1991 network.

## 4 THE SOUND FOLLOWS THE STRUCTURE

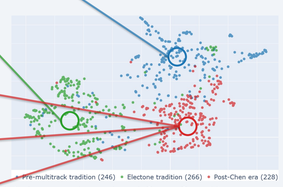
960 recordings (1961–2016). Vocals removed with Hybrid Demucs; instrumental stems embedded with MERT; UMAP projection. MERT was given no dates.




(b) Coloured by arrangement tradition. Three traditions form their own regions; Chen Chih-Yuan's works spread across the whole space.




(a) Coloured by period. Recordings before and after 1978 occupy largely distinct regions.




(c) Recomputed without Chen Chih-Yuan and his protégé. The remaining traditions separate more sharply.

**The person who crossed all traditions also produced sound that crosses all clusters.**

**If MIR is to diversify its objects and its methods, it must ask not only what algorithms can optimize, but what histories they can restore.**

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